

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T30 • Functions

Questions

Student book • questions with writing space

■ 24 pages

FAMILY MAP

Functions

F01	Interpret function notation and mapping	2 Q 2 marks
F02	Compose functions in the stated order	4 Q 9 marks
F03	Control domain and range	5 Q 11 marks
F04	Construct an inverse function	12 Q 48 marks

16 $f(x) = 9 - \sqrt{x}$ where $x \geq 0$

$g(x) = 4x^2$

(a) Find $f(9)$

(1)

16 $f(x) = 9 - \sqrt{x}$ where $x \geq 0$
 $g(x) = 4x^2$

- (b) Solve $fg(x) < 0$
Show clear algebraic working.

.....
(3)

Question 07

4MA1-1H Q15

MODULAR UNIT 2

4 marks

15 The function f is defined as

$$f: x \mapsto \frac{3x + 1}{x - 2}$$

(a) State the value of x that cannot be included in any domain of the function f

.....
(1)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots$
(3)

Question 31

4MA1-2H Q19 (a)

MODULAR UNIT 2

1 marks

19 The functions f and g are such that

$$f:x \mapsto 5x + 7$$

$$g:x \mapsto \frac{5}{2x-9}$$

(a) State which value of x cannot be included in any domain of g

.....
(1)

Question 32

4MA1-2H Q19 (b)

MODULAR UNIT 2

2 marks

19 The functions f and g are such that

$$f:x \mapsto 5x + 7$$

$$g:x \mapsto \frac{5}{2x-9}$$

(b) Find $fg(4)$

.....
(2)

4 marks

Question 14

4MA1-1H Q17

MODULAR UNIT 2

4 marks

17 The functions g and h are such that

$$g(x) = \frac{11}{2x - 5}$$

$$h(x) = x^2 + 4 \quad x \geq 0$$

(a) What value of x must be excluded from any domain of g ?

.....
(1)

(b) Solve $gh(x) = 1$

.....
(3)

Question 17

4MA1-1H Q25 M01

MODULAR UNIT 2

4 marks

25 The function f is such that $f(x) = 2x^2 - 24x + 7$ where $x \geq 6$

Find the inverse function $f^{-1}(x)$

$f^{-1}(x) = \dots\dots\dots$

Question 13

4MA1-1H Q14 Q14(a)

MODULAR UNIT 2

1 marks

14 The function f is defined as

$$f: x \mapsto \frac{2x}{x-6} \quad x \neq 6$$

(a) Find $f(10)$

(1)

Question 14

4MA1-1H Q14 Q14(b)

MODULAR UNIT 2

3 marks

14 The function f is defined as

$$f: x \mapsto \frac{2x}{x-6} \quad x \neq 6$$

(b) Express the inverse function f^{-1} in the form $f^{-1}: x \mapsto \dots$

$$f^{-1}: x \mapsto \dots \dots \dots (3)$$

Question 36

4MA1-2H Q25 Q25(a)

MODULAR UNIT 2

4 marks

25 The function g is defined as

$$g:x \mapsto 5 + 6x - x^2 \quad \text{with domain } \{x:x \geq 3\}$$

(a) Express the inverse function g^{-1} in the form $g^{-1}:x \mapsto \dots$

$$g^{-1}:x \mapsto \dots$$

(4)

Question 37

4MA1-2H Q25 Q25(b)

MODULAR UNIT 2

1 marks

(b) State the domain of g^{-1}

(1)

$$f(x) = x^2 - 2x \qquad g(x) = x + 3$$

Express the inverse function $h^{-1}(x)$ in the form $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots\dots\dots$$

Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$$f^{-1}(x) = \dots\dots\dots$$

$$f(x) = 5x^2 - 10x + 7 \quad \text{where } x \geq 1$$

$$g(x) = 7x - 6$$

(2)

$$f^{-1}(x) = \dots \quad (4)$$

$$f(x) = 5x^2 - 10x + 7 \quad \text{where } x \geq 1$$

$$g(x) = 7x - 6$$

(2)

$$f^{-1}(x) = \dots\dots\dots (4)$$

15 The functions f and g are such that

$$f(x) = 2x - 3$$

$$g(x) = \frac{x}{3x + 1}$$

(a) State the value of x that cannot be included in any domain of g

.....
(1)

(b) Find $gf(x)$
Simplify your answer.

$gf(x) =$
(2)

(c) Express the inverse function g^{-1} in the form $g^{-1}(x) = \dots$

$g^{-1}(x) =$
(3)

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$$g(x) = \frac{x}{3x + 1}$$

(a) State the value of x that cannot be included in any domain of g

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(b) Find $gf(x)$
Simplify your answer.

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(2)

(c) Express the inverse function g^{-1} in the form $g^{-1}(x) = \dots$

$g^{-1}(x) =$
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$$f(x) = 2x - 3$$

$$g(x) = \frac{x}{3x + 1}$$

(a) State the value of x that cannot be included in any domain of g

.....
(1)

(b) Find $gf(x)$
Simplify your answer.

$gf(x) =$
(2)

(c) Express the inverse function g^{-1} in the form $g^{-1}(x) = \dots$

$g^{-1}(x) =$
(3)

(a) Express $5 + 6x - x^2$ in the form $p - (x - q)^2$ where p and q are constants.

(2)

(b) Using your answer to part (a), find the range of values of x for which $f^{-1}(x)$ is positive.

18 The function f is such that $f(x) = \frac{k}{x}$ where $x \neq 0$ and k is an integer.

(a) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$$f^{-1}(x) = \dots \quad (1)$$

The function g is such that $g(x) = 2 - 3x^4$ where $x \neq 0$

The function h is such that $h(x) = \frac{3x}{2-x}$ where $x \neq 2$

(b) (i) Find $g(-2)$

$$\dots \quad (1)$$

(ii) Express the composite function hg in the form $hg(x) = \dots$
Give your answer in its simplest form.

$$hg(x) = \dots \quad (2)$$

25 The function f is such that $f(x) = 3x^2 - 12x + 7$ where $x \leq 2$

Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots\dots\dots$

19 The functions f and g are such that

$$f(x) = 3x - 2$$

$$g(x) = \frac{x}{2x - 1}$$

(a) Find $g(3)$

.....
(1)

(b) Find $gf(x)$

Give your answer in its simplest form.

$gf(x) =$
(2)

(Total for Question 19 is 3 marks)